

COCHABAMBA, Bolivia

WMO: 852230

Lat: 17.421S

Lon: 66.177W

Elev: 8360

StdP: 10.77

Time Zone: -4.00 (W04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	35.7	38.8	15.7	16.7	72.8	20.9	21.4	66.9	20.3	65.4	17.5	67.7	0.2	300	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	26.3	86.3	58.3	84.5	57.9	82.8	57.5	63.0	78.7	62.1	77.6	61.2	76.4	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	101.5	67.5	57.5	96.7	65.2	56.9	94.9	64.6	33.5	78.8	32.7	78.1	31.9	76.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.6	12.7	DB	30.8	90.5	2.4	1.7	29.1	91.7	27.7	92.7	26.3	93.7	24.6	94.9	(4)
(5)			WB	27.5	66.7	1.8	3.3	26.2	69.0	25.2	70.9	24.2	72.8	22.8	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.2	66.4	66.1	66.2	65.0	61.2	58.1	58.3	61.0	64.5	67.5	68.7	68.0	(6)	
	DBStd	4.59	3.49	3.27	3.24	2.67	2.82	2.66	2.45	2.51	2.63	2.60	3.04	3.45	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	833	24	23	25	32	122	206	209	124	38	9	8	13	(9)	
	CDD50	5199	510	452	502	449	347	244	256	342	436	541	562	558	(10)	
	CDD65	558	69	55	62	31	3	0	0	2	25	85	120	106	(11)	
	CDH74	8004	499	421	584	700	537	321	323	593	798	1114	1213	901	(12)	
	CDH80	1401	81	62	95	104	51	18	17	77	138	249	310	199	(13)	
(14) Wind	WSAvg	3.6	3.9	3.4	3.1	2.9	2.4	2.0	2.6	3.4	4.6	4.9	5.0	4.7	(14)	
(15) Precipitation	PrecAvg	18.6	4.8	3.6	2.7	1.0	0.2	0.0	0.0	0.2	0.3	0.6	1.6	3.7	(15)	
	PrecMax	27.8	8.9	10.6	6.5	3.0	1.2	0.4	0.4	1.3	1.4	2.0	4.4	7.8	(16)	
	PrecMin	10.6	1.1	0.6	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	1.4	(17)	
	PrecStd	4.1	2.2	2.0	1.6	0.8	0.3	0.1	0.1	0.3	0.4	0.5	1.0	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.0	85.1	85.9	85.1	83.8	82.2	82.1	84.5	86.2	87.8	88.3	87.8	(19)	
		MCWB	61.6	61.4	60.9	57.8	54.6	53.3	52.4	53.4	55.7	56.7	57.9	59.7	(20)	
	2%	DB	82.3	82.0	82.6	82.7	81.0	80.2	80.1	82.3	83.8	84.6	86.1	84.5	(21)	
		MCWB	59.9	60.6	60.1	57.7	54.6	52.7	52.0	53.3	55.8	57.5	58.3	59.4	(22)	
	5%	DB	79.1	78.9	80.3	80.7	79.2	77.5	77.4	80.2	81.0	82.7	83.9	82.3	(23)	
		MCWB	59.2	59.9	59.9	57.8	54.2	52.3	51.9	53.3	55.5	57.2	58.3	59.1	(24)	
	10%	DB	76.7	76.6	77.2	78.6	77.2	75.4	75.3	77.3	78.9	80.6	80.9	79.1	(25)	
		MCWB	59.1	59.8	59.4	57.3	54.1	52.2	52.0	53.0	55.3	56.9	58.1	58.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	64.5	63.8	62.5	59.3	57.3	56.8	58.4	60.5	62.2	63.4	63.8	(27)	
		MCDB	79.6	79.9	79.2	77.6	76.8	75.1	76.1	77.8	78.9	80.7	81.1	80.4	(28)	
	2%	WB	62.3	62.6	62.2	60.9	57.6	55.6	55.3	56.6	58.8	60.6	61.8	62.1	(29)	
		MCDB	76.4	76.5	77.6	77.0	76.1	75.4	74.5	76.7	78.5	78.7	79.5	78.7	(30)	
	5%	WB	61.0	61.3	61.0	59.5	56.5	54.3	54.0	55.3	57.4	59.2	60.6	61.0	(31)	
		MCDB	74.7	74.6	76.1	76.0	75.1	73.9	73.4	75.8	77.0	77.9	78.0	77.1	(32)	
	10%	WB	59.9	60.4	59.9	58.5	55.3	53.0	52.8	54.1	56.2	58.1	59.3	59.9	(33)	
		MCDB	72.5	72.8	73.9	74.8	73.5	72.1	71.7	74.1	75.4	76.3	76.5	75.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.4	20.7	22.9	29.1	35.2	37.8	37.3	35.8	31.3	28.3	26.3	23.4	(35)	
		MCDBR	25.2	25.5	27.7	32.9	37.7	40.5	40.4	39.7	34.9	31.9	30.9	28.0	(36)	
	5% WB	MCWBR	8.3	8.9	9.4	11.7	14.9	17.4	17.4	15.7	13.1	10.9	10.2	9.1	(37)	
		MCWBR	21.8	22.1	24.5	28.6	33.7	37.4	37.3	35.8	31.9	28.3	26.5	24.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.341	0.342	0.329	0.308	0.297	0.292	0.299	0.368	0.454	0.409	0.359	0.356	(40)	
		taud	2.492	2.500	2.543	2.589	2.553	2.584	2.523	2.273	2.017	2.206	2.377	2.420	(41)	
	Ebn at Noon	Ebn at Noon	318	314	312	307	299	296	296	283	269	291	310	313	(42)	
		Edh at Noon	37	36	34	30	29	27	30	41	56	48	41	40	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1688	1652	1679	1663	1581	1488	1538	1691	1782	1853	1943	1764	(44)	
	RadStd	RadStd	124	95	106	99	75	75	75	84	129	118	108	87	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	+1.28	-5.58	N/A	N/A	N/A	N/A	-148	+196	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page